Wei Wang (王尉)

Ph.D. Candidate in Mathematics, School of Mathematical Sciences, Peking University

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Education

Sept. 2022 – 2027 (expected) **Peking University, School of Mathematical Sciences**
Ph.D. Candidate in Mathematics
Beijing, China
Advisor: Prof. Zhifei Zhang

Sept. 2018 – June 2022 **Zhejiang University, School of Mathematical Sciences**
B.S. in Mathematics and Applied Mathematics
Hangzhou, Zhejiang, China
Advisor: Prof. Ting Zhang
Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

Research Interests

Geometric variational problems; elliptic and parabolic partial differential equations; harmonic maps and their flows; liquid crystal theory; homogenization theory; nonlocal operators.

Publications and Accepted Papers

1. **Wei Wang**. Sharp convergence rates for parabolic Green's functions in time-independent periodic homogenization. Accepted for publication in *Frontiers of Mathematics*. [arXiv: 2606.12101]
2. **Wei Wang** and Zhifei Zhang. Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers. Accepted for publication in *Memoirs of the European Mathematical Society*. [arXiv: 2404.00677]
3. **Wei Wang**, Wei Wang (王伟), and Zhifei Zhang. Existence of weak solutions for two-phase matrix-valued harmonic map flows. *Journal of Functional Analysis* 291 (2026), no. 11, Article No. 111623. [doi] [arXiv: 2506.16278]
4. Haotong Fu, Huaijie Wang, and **Wei Wang**. Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects. *SIAM Journal on Mathematical Analysis* 58 (2026), no. 4, 3306–3324. [doi] [arXiv: 2508.01811]
5. Haotong Fu, **Wei Wang**, Ke Wu, and Zhifei Zhang. Stratification and rectifiability of harmonic map flows via tangent measures. *Mathematische Annalen* 395 (2026), Article No. 92. [doi] [arXiv: 2504.14880]
6. Rupert L. Frank, Tianling Jin, and **Wei Wang**. On the sharp constants in the regional fractional Sobolev inequalities. *SN Partial Differential Equations and Applications* 6 (2025), Article No. 15. [doi] [arXiv: 2403.00357]
7. **Wei Wang**. Uniform estimates of resolvents in homogenization theory of elliptic systems. *Journal of Differential Equations* 370 (2023), 1–65. [doi] [arXiv: 2211.14614]
8. **Wei Wang** and Ting Zhang. Homogenization theory of elliptic systems with lower order terms in dimension two. *Communications on Pure and Applied Analysis* 22 (2023), no. 3, 787–824. [doi] [arXiv: 2210.06329]

Projects

Oct. 2026 **Harmonic Maps in Lean**
Developed two linked Lean 4/Mathlib libraries for small-energy regularity and singular-set stratification of minimizing harmonic maps.
Supplied mathematical proofs and designed reusable theorem interfaces; directed the dimension-independent Luckhaus induction and reviewed assumptions, quantifiers and shared witnesses during AI-assisted implementation.
Project article Regularity code Stratification code

Preprints and Manuscripts

1. Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Asymptotics for two-dimensional parabolic vectorial Allen–Cahn systems. Preprint. [arXiv: 2609.27596]
2. Haotong Fu, Kelei Wang, **Wei Wang**, and Ke Wu. Small-energy regularity for stationary weak solutions of the Liouville equation. Preprint. [arXiv: 2609.26026]
3. Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Boundary asymptotics for two-dimensional vectorial Allen–Cahn systems. Preprint. [arXiv: 2609.00739]
4. Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Boundary regularity for Oseen–Frank minimizers with arbitrary positive splay, twist, and bend constants. Preprint. [arXiv: 2608.22287]
5. **Wei Wang** and Qin Wu. Sharp instability of planar isotropic–nematic interfaces in the Landau–de Gennes model. Preprint. [arXiv: 2608.12131]
6. Haotong Fu, Huaijie Wang, and **Wei Wang**. Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime. Preprint. [arXiv: 2608.08456]
7. Haotong Fu, Huaijie Wang, **Wei Wang**, and Zhifei Zhang. On the absence of point defects in biaxial Landau–de Gennes models. Preprint. [arXiv: 2608.04908]
8. Giacomo Canevari, Haotong Fu, and **Wei Wang**. Singular limits for three-dimensional global minimizers of Ginzburg–Landau-type functionals: uniform estimates and singular sets. Preprint. [arXiv: 2606.27691]
9. Haotong Fu and **Wei Wang**. A note on an approximate fixed point theorem for VMO maps. Preprint. [arXiv: 2606.15670]
10. Giacomo Canevari, Haotong Fu, and **Wei Wang**. Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions. Preprint. [arXiv: 2605.04442]
11. Haotong Fu, Huaijie Wang, and **Wei Wang**. Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit. Preprint. [arXiv: 2507.14955]
12. **Wei Wang** and Zhifei Zhang. Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity. Preprint. [arXiv: 2411.16048]
13. Haotong Fu, **Wei Wang**, and Zhifei Zhang. Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations. Preprint. [arXiv: 2408.06726]

Research Visits and Programs

Mar. – Apr. 2026 — **University of Verona, Department of Computer Science**
Visiting Ph.D. Student
Verona, Italy
Host: Prof. Giacomo Canevari

June – Sept. 2021 — **HKUST, Department of Mathematics**
Summer Research Program (Online)
Hong Kong, China
Advisor: Prof. Tianling Jin

Selected Talks

Aug. 2026 — Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals: Uniform Estimates and Singular Sets
Conference on Geometric Measure Theory and Partial Differential Equations
Enshi, Hubei, China.

July 2026 — Uniform Estimates for Landau–de Gennes Minimizers with Point and Line Defects
Academic Forum on Theory and Numerical Analysis of Liquid Crystals and Complex Systems
Xiangtan University.

Jan. 2026 — Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers
Academic Workshop on Fluid Dynamics Equations
Peking University.

Nov. 2025 Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity
Jinhua Conference on Harmonic Analysis and Its Applications
Zhejiang Normal University.

Aug. 2025 Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit
Conference on Theory and Computation of Quasi Crystals
Nankai University.

June 2025 Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures
PDE and Geometry Seminar
Zhejiang University.

Grants and Projects

2023 – 2028 **Physical Modelling and Mathematical Theory of Liquid Crystals**
Participant, NSFC, Grant No. 12288101.

2023 – 2028 **Physical Modeling and Mathematical Theory of Liquid Crystal Systems**
Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

Academic Service

- **Mathematical Reviews / MathSciNet, American Mathematical Society**
  Registered prospective reviewer.
- **Academic Committee, 4th Frontiers Doctoral Forum**
  Screening and organization of talks in Analysis and Differential Equations.

Teaching Experience

2022 – 2026 **Peking University, Teaching Assistant**
Advanced Mathematics (A, B, and C); Real Analysis; Functional Analysis (I); Harmonic Analysis.

2021 – 2022 **Zhejiang University, Teaching Assistant**
Mathematical Analysis (I and II).

Honors and Awards

Sept. 2026 **National Scholarship**, School of Mathematical Sciences, Peking University.

Apr. 2025 **First Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.

Apr. 2024 **Third Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.

Oct. 2023 **Elite Program for Graduate Students in Mathematics**, Peking University.

Aug. 2022 **Nomination Award, ICCM Creative Undergraduate Thesis**, Shing-Tung Yau Center of Southeast University.

Apr. 2021 **First Prize, 11th Chinese Mathematics Competitions Finals**, Wuhan University.

June 2021 **Nomination Award, S.-T. Yau College Student Mathematics Contest**, Yau Mathematical Sciences Center, Tsinghua University.

Skills and Languages

Skills LaTeX, MATLAB, Mathematica, Python, Lean.

Languages Chinese (native), English (fluent).